Lead 16.6: the affine saturation law on a genus-2 Jacobian

Round 4 of `RESEARCH_ISOGENY_FIBERS_ECDLP.md`, 2026-10-11. Code: `examples/r4_16_6_genus2_census.rs`. Raw records: `research/isogeny_fibers_20261010/round4/16_6_genus2_law/raw/`.

1. Question and verdict

Question (§16.6). Prop. 15.2 prices factor-base symmetry for the mitm pipeline as cost factor $|M|^{m-2}\ /(|A|\cdot i)$. Does that hold on the Jacobian of $y^2 = x^5 + b$ over $\mathbb{F}_p$ ($p \equiv 1 \pmod{10}$), where the automorphism group $A = \langle \zeta_5, \iota \rangle$ has order 10 and rational (2,2)-kernels exist? **Falsifier:** a measured factor outside the law beyond sampling error.

Verdict: confirmed, with one structural difference.

1. All arms follow the law (measured, $p = 4111$; predicted in brackets). The m = 3 cost ratios are: A-closed 0.198–0.200 [0.2], μ = 2 1.89–1.98 [2], Richelot μ = 4 3.82 [4], non-isotropic μ = 4 3.82–3.92 [4], μ = 5 4.85 [5], combined A + M_5 0.977 [1]. The class factors are 1.00, 0.255, 0.064 and 0.041 [1, 0.25, 0.0625, 0.04]. The image arms reach c_2 ratio 16.0 [16] and 5.02 [5].
2. One leading-order deviation, at m = 2 on the A-closed base when A fixes the cofactor classes: measured +32%, +21% and +9% at $|F| = 50, 100, 200$. It matches the derived structured-pair term $1 + (h-1)(|A|-2)/(2|F|)$ = 1.32, 1.16 and 1.08. That term vanishes as $1/\ |F|$ and has the same form for $j = 0$ elliptic curves, so it is not a new regime.
3. Genus 2 differs in **realizability**. A base of degree-1 divisors is never closed under a nonzero translation, because $|(C - \infty) \cap (C - \infty + t)| \le 2$. Kernel-saturated bases must therefore be $(\mu - 1)/\mu$ degree-2 divisors. That is fine for the mitm census, but unusable by a Gaudry/Nagao oracle. Automorphism folding stays free.

2. Setup

Two families, three primes each, all validated: the exhaustive Mumford count equals $L(1)$, $[\#J]D = 0$, associativity holds, Cantor agrees with `jv_cover::Hyp`, and ζ acts on G as a 5th root of unity mod r. **C2** has $-b$ a 5th power, so $J[2] \subset J(\mathbb{F}_p)$ and $h = 16$: $p = 1051, 2131, 4111$ with $b = 1042, 32, 2172$. **C1** has b square, $T_5 = \left(0, \sqrt{b}\right) - \infty$ and $h = 5$: $b = 49, 4, 3984$. The census is the one of §15.2. D_m is the exact number of distinct nonzero classes of G that are m-sums of base elements. Columns are orbits of $[h]F$ under A. The cost is $U \cdot r/D_m \cdot L_m$. Each cell has 40 random bases; the figures show ratios of means with 95% bootstrap intervals. Costs use the plain base's nominal $|F|\ /2$ columns, because the ζ-stable x-set folds a few plain pairs by accident (plain U = 0.83 / 0.92 / 0.95 of nominal).

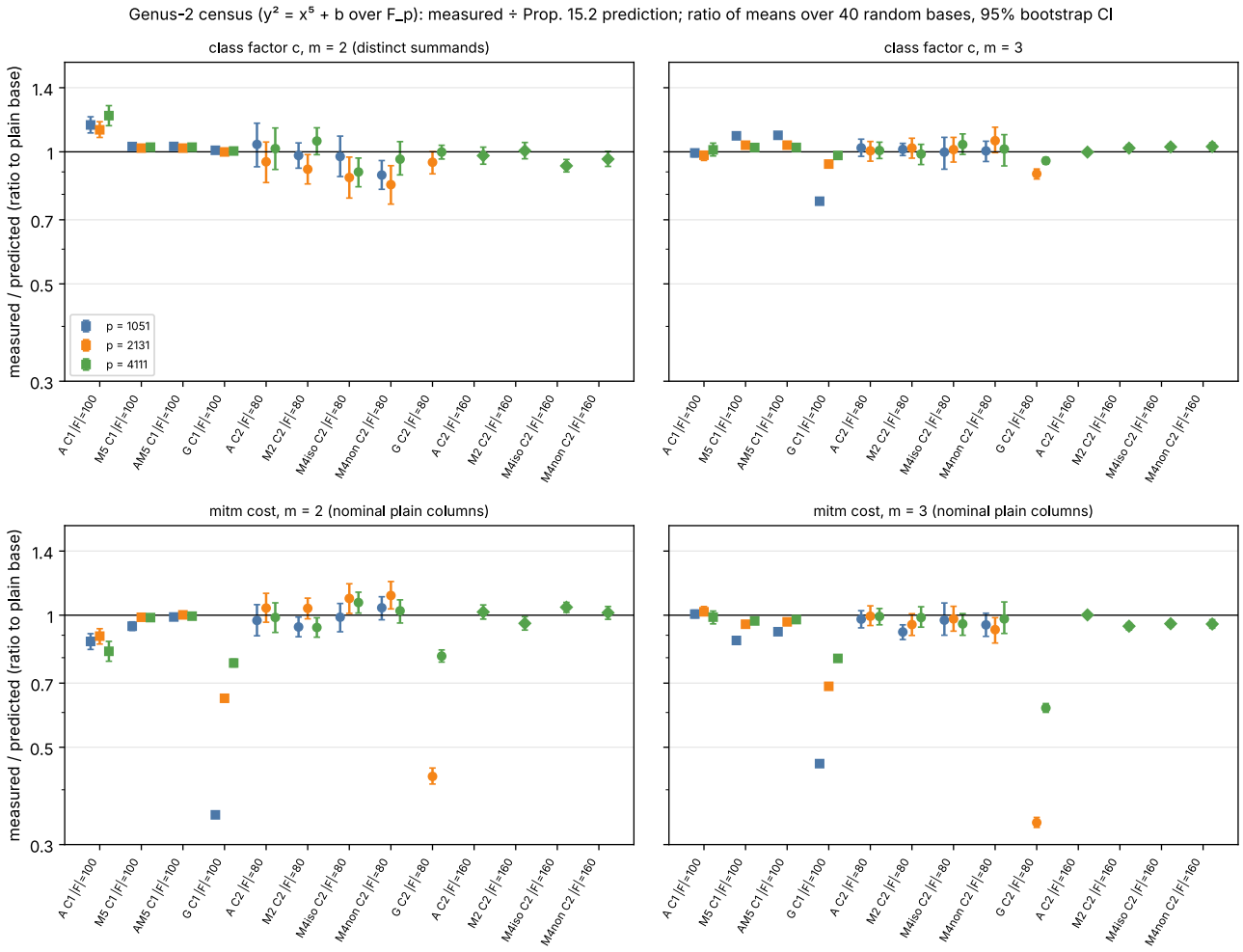


Figure 1: Measured ÷ predicted for every arm and curve (1 = exactly the law). Points: ratio of means over 40 random bases; bars: 95% bootstrap CI; colour = p. The G arms fall below 1 in cost because the G-filtered base is partly A-closed by itself (U ratio 0.32–0.82), and their c_3 falls below 1 at small r (hit rate up to 0.54). Records: `raw/census_*.json`, data `plot_data.json`.

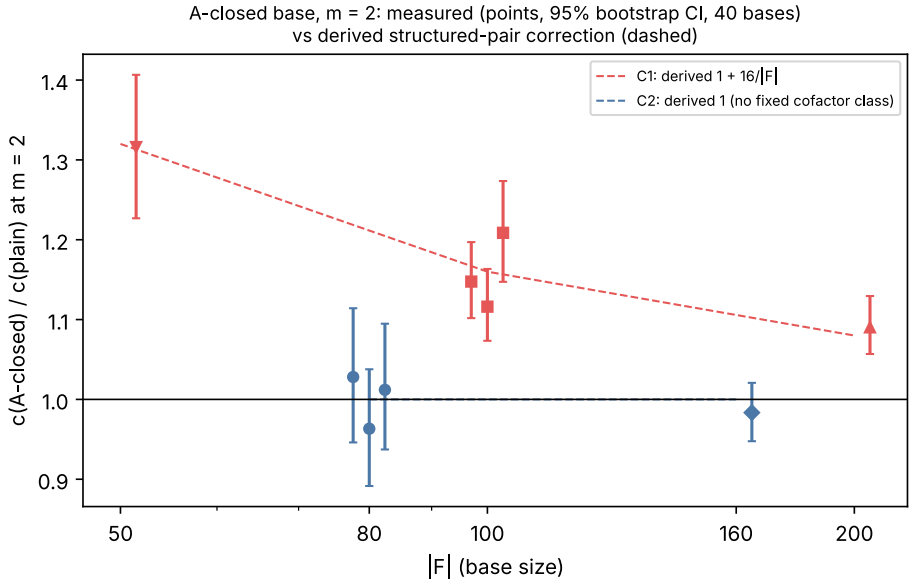


Figure 2: A-closed base, class-factor ratio at m = 2 against $|F|$. On C1 (A fixes $H = \langle T_5 \rangle$) the pairs $\{\zeta^a P, \zeta^b(-P)\}$ always land in G, which gives the derived term $1 + 16/|F|$ (dashed; `finite_size_A.json`). On C2 ζ fixes no nonzero cofactor class and the ratio is 1. Measured: `raw/census_c1_p4111_s{50,200}_Aonly.json` and the main runs.

3. Key measured values (p = 4111, ratio to plain, [prediction])

curve, |F|	arm	U	c, m = 2 strict	c, m = 3	cost m = 2	cost m = 3
C2, 160	A	0.217 [0.2]	0.98 [1]	0.999 [1]	0.203 [0.2]	0.200 [0.2]
C2, 160	M2	0.523 [0.5]	0.503 [0.5]	0.255 [0.25]	0.958 [1]	1.888 [2]
C2, 160	M4 Richelot	0.266 [0.25]	0.232 [0.25]	0.064 [0.0625]	1.043 [1]	3.825 [4]
C2, 160	M4 non-iso.	0.266 [0.25]	0.241 [0.25]	0.064 [0.0625]	1.013 [1]	3.819 [4]
C2, 80	G, i = 16	0.611 [1]	15.98 [16]	15.27 [16]	0.050 [0.0625]	0.038 [0.0625]
C1, 100	A	0.211 [0.2]	1.209 [1.16 corr.]	1.010 [1]	0.165 [0.172 corr.]	0.198 [0.2]
C1, 100	M5	0.209 [0.2]	0.205 [0.2]	0.041 [0.04]	0.987 [1]	4.851 [5]
C1, 100	A + M5	0.042 [0.04]	0.205 [0.2]	0.041 [0.04]	0.199 [0.2]	0.977 [1]
C1, 100	G, i = 5	0.824 [1]	5.022 [5]	4.906 [5]	0.156 [0.2]	0.159 [0.2]

Intervals for every cell and the p = 1051 and 2131 runs are in `RESULTS.md` and `plot_data.json`. The saturated m = 3 costs climb toward the prediction as r grows (M5: 4.38 $\longrightarrow$ 4.77 $\longrightarrow$ 4.85), as in §15.2.

4. Limits

Toy sizes ($p \leq 4111$, $r \leq 3.3 \cdot 10^6$, $|F| \leq 200$); the mitm cost **model**, not solves; no algebraic oracle; no explicit Richelot image arm (the image arm is $[2]J = G$ on C2). There is no claim about any deployed curve and no sub-rho claim. Wall time was not used. The commands are in `run_all.sh`.